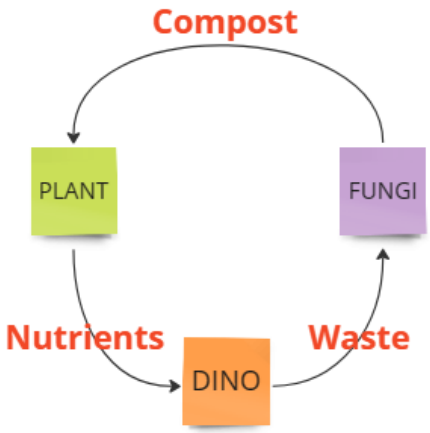
🍄 Fungi

Growth/Propagation (VISUAL polish for Beta - low priority)

Proposed changes:

- The fungi never shrink or die, as this goes against the idea of building towards lushness.
  - “Maximum Lifetime Compost Produced” is reused as *maximum growth*.
- The fungi keep growing as it currently does, but at every “Propagate Growth Percent” it propagates a new instance of the model (not the blueprint).
  - For example, at 0.25 it will propagate 4 times - at 0.25, 0.50, 0.75 and 1.
  - The propagated model starts at the smallest scale and grows similar to its parent but has a “Propagated Max Scale,” which caps its maximum size. If set to 0.8, it grows to be 80 percent of the size of its parent.
    - “Propagated Random Scale” adjusts the random range. If set to 0.1 and the number above is 0.8, the max scale will be a random range between 0.7 and 0.9.
- “Propagated Min Distance” and Propagated Max Distance” define the random range for how far away from the parent it grows. Direction is random, but if the direction collides with a collider within the random range, then it chooses a new random direction.
- This propagated model is part of the scan result for the original BP/model/instance and does not contribute to the Harmony; it’s just a visual representation of growth and progression (lushness).
  - The propagated model will appear as part of the parent BP, so the scannable point and the scan result will be for the parent BP.
  - The propagated model may not propagate more fungi.
- “Propageted Range Increase” increases the range of the area-of-effect for each propagated unit.
  - For example, if set to 0.1, the area-of-effect radius increases by 10 percent for each propagated unit.



Key Creative Details

- Fungi are also placeable objects that can be placed/planted from the inventory for a cost
- Player can influence the efficiency of the ecosystem without having to control the ecosystem itself
  - Functionally a hybrid of Plants and Dinos (has hunger, eats poo (waste))
  - Eats waste left nearby
    - Fungi “nearby” Dung will remove the stored waste Points in the Dung (that would otherwise also be added to the waste Points bar)
  - Players are encouraged to plant fungi in patches, as fungi has a multiplier if multiple mushrooms are close together (Generates extra nutrient value from waste)
  - Mycelium network breaks dung down based on proximity. The closer to the fungus, the faster it is broken down.
  - Fungi acts as a silo for Compost and dies as those Compost points are all consumed, disappearing when empty. They can produce X Compost over their lifetime and die when they hit their limit and are empty of Compost.
  - Fungi harmony is calculated by comparing the number of fungi against the number of plants.
  - There must be 1 Fungi connected to every group of three plants.

Overview

- [ ] Hero Fungi are environment object placed by the player.
  - [ ] They are purchased using Dino Points and planted using the planter.
- [ ] Generates Compost points over time. See Compost Generation.
  - [ ] Has a max lifetime generation
  - [ ] Lifetime of X units
  - [ ] Shrinks as it does when it approaches this ceiling

- [ ] Storage of X unit
  - [ ] Has a tick
  - [ ] Gives compost every tick
- [ ] Can grow and change scale.
  - [ ] Growing rate can be accelerated.
  - [ ] Growth tied to Compost generation
- [ ] Have an outer detection radius used to sense plants, fungi and waste.
  - [ ] Have an inner detection radius used to sense 'close' fungi
  - [ ] Multiply Compost production when close to X number of other fungi
  - [ ] Multiply waste disposal rate when it's exceptionally close
  - [ ] Compost can be transferred to plants inside the detection radius
- [ ] Dinosaurs 'prefer' to use the area covered by mycelium to defacate
- [ ] Has a cap for the number of plants it can serve

Purpose

Plants produce nutrients which Dinosaurs eat, and dinosaurs produce waste.
The purpose of the Fungi portion of the Harmony System is to take the waste produced by dinosaurs and transfer it into a useable resource for plants and complete the circle.

Fungi decompose waste in the environment and convert it to Compost which is then held in the mushroom like a silo. Plants that inside the detection radius of the fungi convert these points to nutrients.

Waste

Dino's naturally produce dung. Dinos will produce dung at a configurable rate but by default the rate is equivalent to the number of hunger points shed.

The dung will appear around the map but will have a configurable chance to appear inside the detection radius of any fungi near plants.

Similar to how Plants store Nutrients within themselves, **Dung will have a set XXX waste Points when they spawn**.

Fungi assets will have a radius (visible to the player when planting fungi) and if that radius overlaps other fungi, then it would be considered "nearby".

The **waste points stored by the Dung** will be reduced by Fungi at a set Waste Point Removal Rate. Dung completely disappears from the environment when its Waste Points hits 0.

Harmony is checked by comparing the total Compost Points stored by Fungi versus the maximum nutrient capacity of all plants in the environment, and a configurable percentage is used as the metric for passing.

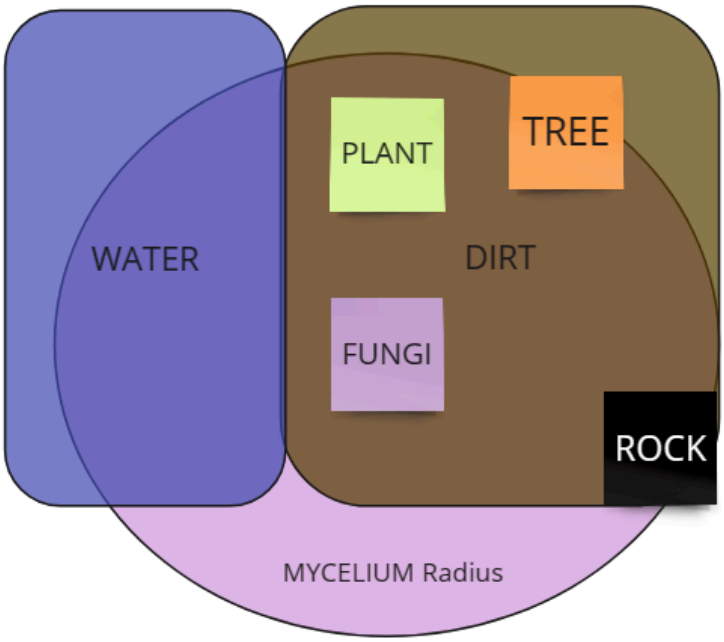
For more information on how waste works, refer to [the waste page subsection]() of the dino.

Planting & Status

Hero Fungus is placed by the player using the [planter]().
An exception may be the tutorial in which designers place 'starter' fungi and the player must scan these to unlock the first fungi type.
Fungi cost a configurable amount of Dino Points to plant (**Fungi Cost DP)** .
When fungi are removed, a percentage of these are reclaimed and this is declared as the (**Fungi Removal DP)**.

Fungi placed by the player has a purple glow when the scanner is held.
Environmental fungi (if they exist) have no glow and are not interactable.

Fungi have a detection radius. This can be visualized when "planting" as a circle around the fungi.
The player can also review this radius by scanning a fungus after it is already planted.
This radius represents the underground mycelium network of the fungus.

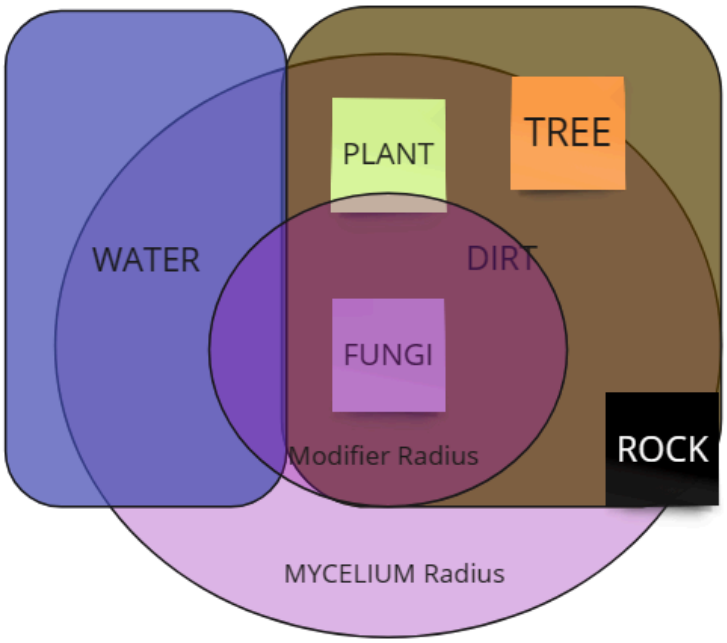


The plant and the tree are able to make use of Compost generated by this fungi.

Compost

Fungi have a stored **Compost Value** and a **Compost Production Rate**.
Compost is created by converting Waste at a variable rate.
The rate is modifiable by proximity to other fungi and distance to waste.
The ratio of conversion from waste to Compost should have an adjustable base, set it to 1:1 by default.

Players can improve this conversion rate by planting fungi close together (inside the inner detection radius) and by overlapping plants within the outer detection radius.

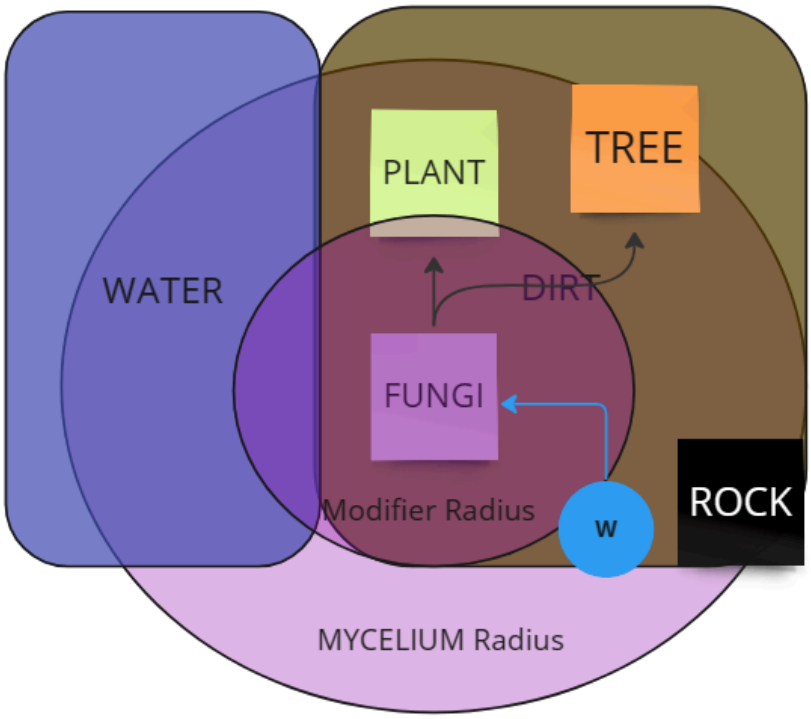


Fungi planted inside the inner radius add a multiplier to Compost prodiction. The outer radius affects Compost transfer.

As the fungi generates Compost, it increases in the mushrooms physical size (scales up).
The bigger the fungi, the more Compost it holds. There is a configurable cap on the amount it can hold.
Compost is visualized by small shoots of fungi on the ground around the parent (E.G. If it holds 100 Compost Points, a small, mushroom appears every 10 points.)

Compost points stored in fungi can be read/ accessed by scanning the fungi using the scanner. The data that is scanned will populate the Scan Results Card which will have a link to the DinoDex.

The player receives periodic updates about the points being generated by their fungi, these are visualized on the UI.



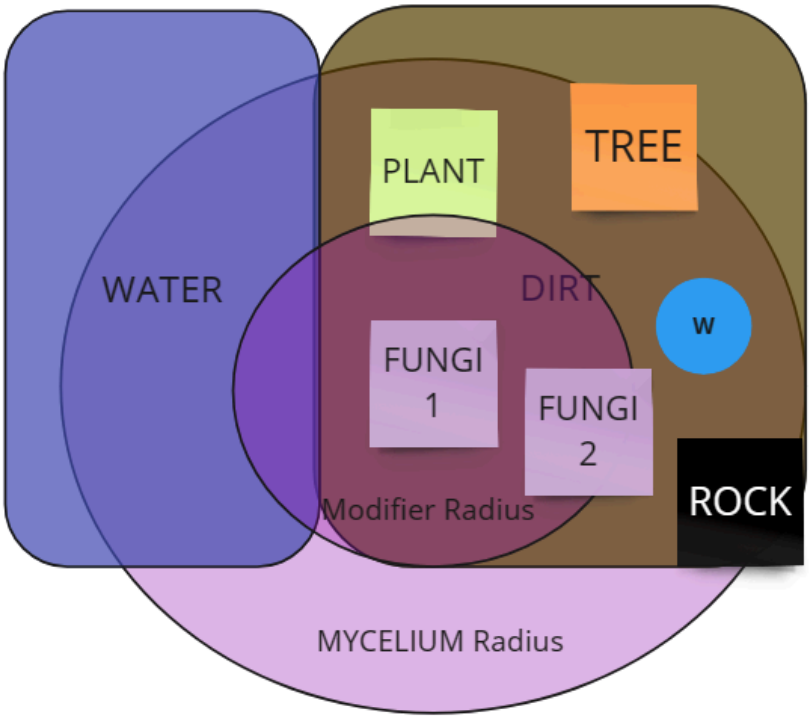
The Fungus in this diagram is taking *waste* (blue circle) and transforming it to *Compost* which it distributes through the mycelium to the Tree and The Plant.

If fungi are not producing enough resources to support the plants, the player should see the issue reflected in the slow growth of their plants and so in the reduced health of their dinosaurs.

Mycelium Networking

Fungi receives a multiplier to Compost generation when their inner radii closely overlap.
The number of fungi needed to apply this modifier is a configurable variable.
The range of this overlap is an adjustable value.

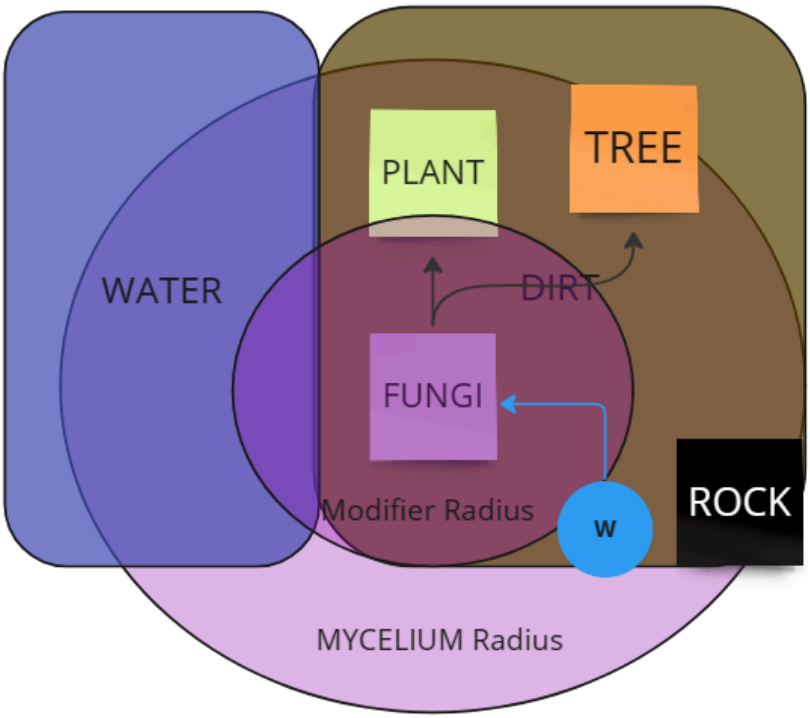
Waste that is located inside the inner radius is also broken down faster.



Fungi 1 and 2 are close enough to receive a bonus to Compost generation.

Mycelium ‘networks’ are used to transfer Compost to plants and other fungi. These are visualized as purple lines when scanned and show the movement of resources live.

Fungi needs to be planted in close enough proximity to plants and mushrooms to transfer the resources - this is determined by the radius. If the radius overlaps, it is part of the network. Plants prefer to use resources from the nearest mushroom when multiple mushrooms are on a network.

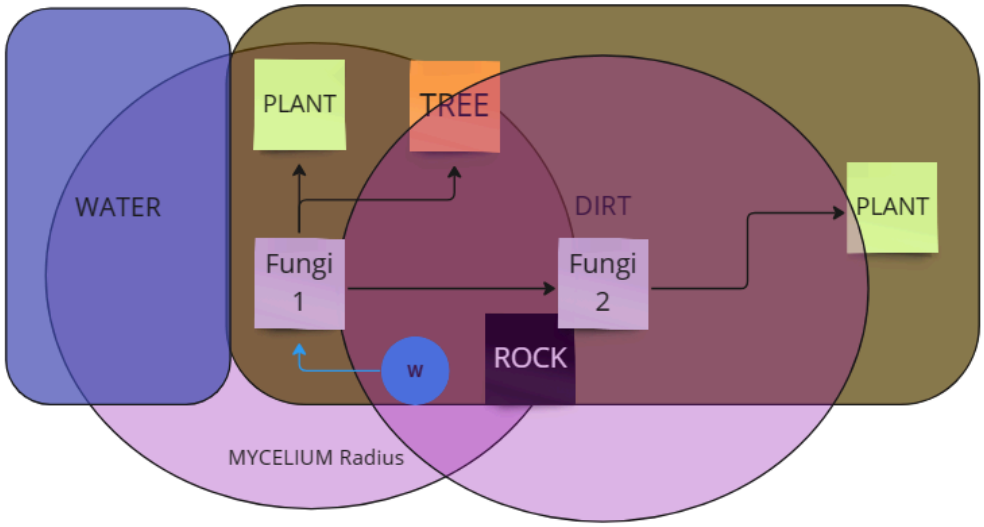


The Fungus in this diagram is taking *waste* (blue circle) and transforming it to *Compost* which it distributes through the mycelium to the Tree and The Plant.

Overlapping mycelium networks can be used to transport Compost to faraway plants.

Visualized when scanned, shows connections between mushrooms and points being transported from one network to another.

Mushroom closest to waste takes the waste points. Overproduction is distributed to the next closest mushroom. If they're the exact same distance it picks one at random. The networks are visualized when scanned - connections between mushrooms are shown as lines and points being transported from one network to another are shown as bright portions moving along these lines.



Fungi Harmony

Alpha:

Fungi Harmony is determined by the ratio of fungi to plants.
For every three plants there must be one unit of fungi in the map.

This means that if you have three ferns and one fungi and one baby Triceratops, all three meters will fill up when the ferns are at full nutrients, the baby has eaten it's fill and the mushrooms are placed.

An expanded example:
1 fungi = 3 plants, then the fungi harmony meter is = 100%
2 fungi (6 plant units) with 6 plants = 100%
1 fungi (3 plant units) with 6 plants = 50%
1 fungi (3 plant units) with 9 plants = 33%
1 fungi (3 plant units) with 12 plants = 25%
3 fungi (9 plant units) with 12 plants = 75%And harmony is calculated the same way, if the bar is 80% or above, it's in harmony

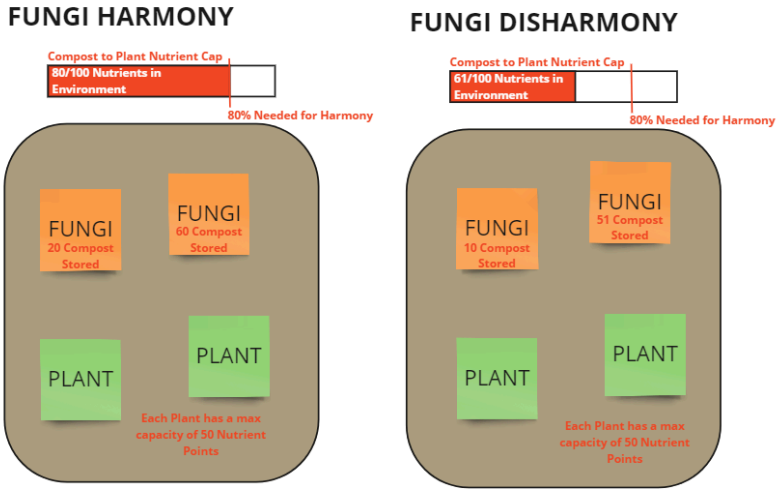
Iterative Post Alpha:
The condition for fungi harmony should be expanded to detect if the plants are being SERVED by all the fungi - this can be achieved by forcing fungi to be planted near plants only.

Variable		
NumberOfPlants	The target number of plants for a grouping.	3
NumberOfFungi	The minimum number of fungi required for each grouping.	1

~~Harmony is checked by comparing the total Compost stored by fungi versus the maximum nutrient capacity of all plants in the environment, and a configurable percentage is used as the metric for passing.~~

~~If the amount of Compost Points in the environment meets a **XX% threshold** of the **total nutrient capacity of plants** the **fungi (purple) section of the Harmony Meter lights up.**~~

~~Once, all three pieces of the Harmony Meter are lit up/at Harmony threshold, is the ecosystem considered in Harmony. And of course, the different parts of the ecosystem can move in and out of Harmony (i.e., the pie chart can light up and un-light).~~



Stretch Goals

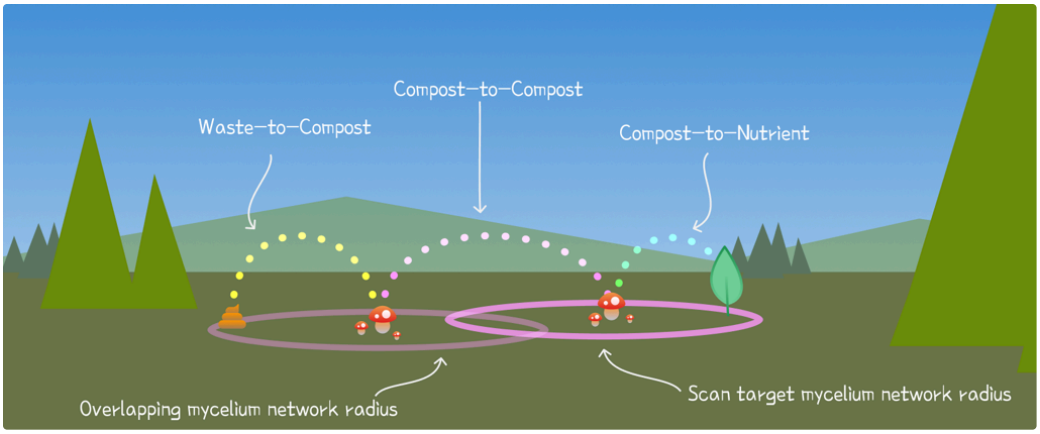
Percent chance to propagate a random plant when a new dinosaur has recently pooped near an uncommitted mushroom.

Variable	Description	Example Value
Mycelium Network Radius	The size of the outer radius used to detect waste and transport Compost to plants.	
Compost Value	The amount of Compost stored in the mushroom.	
Compost Production Rate	The rate at which Compost is produced.	
Compost Bonus Radius	The inner radius - If extra fungi are located here there is a bonus to Compost points generated.	
Fungus for Bonus	The number of nearby fungus that affects production of Compost.	
Compost Bonus Modifier	The modifier applied to the number of nitrogren points produced from one waste point.	
Waste Proximity Breakdown Modifier	How much faster waste is broken down to Compost if inside the Bonus Radius.	
Waste Breakdown Rate	How long it takes to breakdown 1 waste point and make 1 Compost point	2 point per minute = 1 fungi 4 point per minute = 2 fungi
~~Compost To Nutrient Ratio~~	~~The ratio at which Compost becomes~~	

	~~Nutrients~~	
Fungi Cost DP	How much **DP** costs to place the fungi actor into the environment from the inventory.	100
Fungi Removal DP	How much **DP** player recieves for removing fungi. A percentage or portion of the Fungi Cost DP.	
Plant Capacity	How many plants a mushroom can serve.	
Network Size	How many mushrooms are connected to the network.	
Waste Queue Capacity	The max amount of available “slots” that the fungi can have of nearby dung. If more dung is nearby a fungi than a fungi will allow, no conversion effect will happen on the extra Dung.	

Fungi Scan

When fungi are scanned they will show the above described network as a connection between dung to fungi to plants that are within its radius.


Compost-to-Compost
Waste-to-Compost
Compost-to-Nutrient
Overlapping mycelium network radius
Scan target mycelium network radius